Wajid Karim

AI Engineer
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SUMMARY

AI Engineer with hands-on production experience in multi-agent systems, LLM pipelines, and RAG architectures. Proven ability to build and deploy intelligent systems across healthcare, automation, and analytics domains. Background in full ML pipelines from data annotation and model training to deployment and monitoring. Passionate about applying cutting-edge AI to solve real-world problems.

SKILLS

Languages: Python, SQL, JavaScript, HTML/CSS, Bash, PowerShell
Frameworks: Flask, FastAPI, Django, WebSockets, React, Vite
AI / ML: PyTorch, TensorFlow, Keras, scikit-learn, Hugging Face Transformers, OpenAI / Azure OpenAI, LangChain, LangGraph, RAG, LLM Fine-Tuning, Prompt Engineering, Agentic Frameworks, MLflow, Structured Outputs
NLP: NLTK, spaCy, intent classification, NER, entity extraction, word embeddings, text similarity
Vector DBs & LLM Ops: pgvector, hybrid search (dense + sparse), re-ranking pipelines, hallucination detection, faithfulness scoring, token cost optimization, context window management
Cloud & DevOps: Microsoft Azure, AWS, GCP, Docker, Kubernetes, CI/CD, GitHub Actions, Azure Pipelines
Databases: PostgreSQL, MySQL, MongoDB, SQL Server
Computer Vision: OpenCV, MediaPipe, YOLO, DeepSORT
Others: Agile/Scrum, Git, OOP, Analytical Thinking, Problem-Solving, Communication, Teamwork

PROFESSIONAL EXPERIENCE

AI Engineer, *Canvas Solution* 02/2026 – 04/2026

- Built ASA-Agent, a production multi-agent healthcare assistant using LangGraph and LangChain with intent routing across scheduling, emergency, cancellation, and FAQ workflows.
- Designed dual RAG pipeline strategy (FaqNode & FaqNodeNoTool) with optimized chunking strategies, embedding models, and context window management for medical knowledge retrieval.
- Implemented semantic search via pgvector (PostgreSQL) for dense embedding storage and retrieval; applied re-ranking logic to improve answer faithfulness.
- Enforced structured outputs using Pydantic schemas; monitored model performance using MLflow to track inference quality and latency tradeoffs.
- Applied NLP techniques including intent classification and entity extraction to route patient queries across specialized agents.

AI Data Annotator / ML Intern, *Unpeel (Startup)* 12/2025 – 02/2026

- Collected and structured children's fashion product datasets across 19 brands for training ML classification models.
- Labeled and validated product attributes (gender, clothing type, brand) to support NLP-based text classification pipelines.
- Supported the full ML pipeline: data collection to labeling to preprocessing to model training preparation.

Machine Learning / Deep Learning Fellow, *Buildables* 07/2025 – 11/2025

- Completed 3-month intensive fellowship in ML and Deep Learning; worked on research-driven practical AI projects.

- Received strong mentor recommendation for exceptional performance.

Junior Data Scientist & AI Engineer, *IT Directorate* 06/2024 – 12/2024

- Analyzed organizational data using Python and scikit-learn to build models supporting 3+ strategic decisions by department leadership.
- Built predictive dashboards for sales and customer analytics, reducing manual reporting time by ~40% and improving insight turnaround.
- Led AI-driven automation projects using TensorFlow/Keras, eliminating repetitive manual processes across 2 operational workflows.

PROJECTS

RAG Chatbot*, Flask, LangChain, pgvector, OpenAI*

- Production RAG chatbot with optimized chunking, dense vector retrieval via pgvector, and re-ranking for improved semantic accuracy.
- Applied intent classification and entity extraction for query routing; monitored faithfulness scoring and hallucination rates.
- Deployed via Flask API with context window and token budget management for multi-turn conversations.

AI Browser Assistant*, Python, FastAPI, Playwright MCP, React*

- Multi-agent browser automation system with context-aware summarization, smart form filling, and tab control using OpenAI and Gemini models.
- Developed FastAPI backend and two Chrome extensions (Playwright MCP bridge and React side panel).
- Modular architecture combining AI agents, browser automation, and React + Vite frontend stack.

Multi-Agent Supervisor System*, LangGraph, FastAPI, Groq, SQLite*

- Built a sophisticated multi-agent supervisor system using LangGraph and FastAPI, implementing adaptive routing across specialized agents: Researcher (DuckDuckGo), Wikipedia, and Analyst.
- Designed a stateful graph architecture with SQLite-based checkpointing for conversation persistence across sessions and thread-based context management.
- Implemented a REST API via FastAPI (/v1/execute) with full execution trace logging, token usage tracking, and loop prevention for production robustness.
- Integrated Groq LLM for supervisor decision-making; each agent uses custom tools for web search, Wikipedia lookup, and Python-based calculations.

Object Detection for Autonomous Driving*, YOLOv8, Python*

- Deployed YOLOv8s for roadside object detection (vehicles, pedestrians, signs); evaluated with mAP metrics and designed visual analysis tools.

Student Performance Prediction System*, Flask, scikit-learn, Python*

- End-to-end ML system predicting student performance with 82% accuracy; multi-model pipeline with automated hyperparameter tuning.

EDUCATION

Karakoram International University (KIU),
Bachelor of Science in Computer Science

CERTIFICATIONS

- AI and Data Science Certification — **NUST** — 6-month intensive program, 2024: Machine Learning Fundamentals, Python, Data Analysis, Neural Networks, NLP, intent classification, NER
- Microsoft AI & ML Engineering — **Coursera** — Professional Certificate, 2025: Azure ML, intelligent agents, LLM deployment pipelines, context window optimization

- Summer School in Machine Learning — **Cohere Labs** Open Science Community, 2025: NLP, AI Agents, Reinforcement Learning, LLM observability, and mathematical foundations

ADDITIONAL SKILLS

Languages: English (Fluent), Urdu (Native)
Soft Skills: Problem-solving, analytical thinking, self-motivated learner, time management, communication, teamwork